

tously used material. Unfortunately, alloys like these faced the problem of getting accidentally demagnetised, and they were later replaced by ferrite magnets. Despite their relatively lower strength, they were lighter than Alnico magnets, thus allowing the permanent magnet to be larger in size to compensate for the strength.

It was mentioned earlier that the outer rim of the diaphragm is attached to the loudspeaker enclosure. The importance of the enclosure comes into play when it prevents sound waves emanating from the back of the speaker system from interfering destructively with those from the front. Without an enclosure, the waves emitted at the back cancel those emitted forward because they are out of phase, which significantly degrades the level and quality of sound at low frequencies.

The structure of the enclosure should prevent transmission of the sound emitted from the rear of the loudspeaker. This can be made possible by confining the sound in a rigid and airtight box. However, such a structure would reflect sound internally, which will again be thrown back out through the loudspeaker diaphragm. Hence, in addition to the shape, the loudspeaker material should also be taken care of. A good candidate for a



Enclosed loudspeaker system with drivers

speaker system's enclosing material would be one that can absorb sound internally reflected off the speakers. Hence, manufacturers stick to materials like glass wool, medium density fibreboard or synthetic fibre batting. In most high-end loudspeaker systems, specialized speakers are used to produce specific frequency ranges, and the incoming signal is split by a crossover (a sort of electronic filter). These specialised speakers are called drivers, and are generally combined into a complete loudspeaker system to provide performance beyond that constraint.